LOAN APPROVAL PREDICTION

Fundamentals of Data Science
Case Study Report

Presented By:

Team Member's	Name :-	Regd No. :-
1	Shivani Kumari	2401080001
2	Sadhna Kumari	2401080002
3	Ipsita Mohanty	2401080014
4	Goldi Kumari	2401080041
5	Laxmipriya Sethi	2401080063
6	Barsarani Patra	2401080068

Under the Guidance of:

Ms. Priya Mishra & Ms. Pooja Rani Khuntia

DECLARATION

We, the undersigned students— **Shivani Kumari, Sadhna Kumari, Ipsita Mohanty, Goldi Kumari, Laxmipriya Sethi, Barsarani Patra**—enrolled in the Bachelor of Computer Applications (BCA) program at C V Raman Global University, declare that the project titled **"Loan Approval Prediction"** is our original work undertaken as part of the **Fundamentals of Data Science** course for the academic year 2024–2027.

This project aims to design and develop a Loan Approval Prediction system using data science and machine learning techniques. The system analyzes applicant data such as income, credit history, loan amount, and other relevant factors to predict whether a loan should be approved or not. It is developed based on the concepts of data analysis, data preprocessing, and predictive modeling learned during the course.

We attest that the content presented in this project is authentic, and no part of it has been submitted elsewhere for academic purposes.

We acknowledge and take full responsibility for the authenticity, accuracy, and originality of this project work. Any external sources used for reference have been properly cited. We understand the importance of academic integrity and affirm that this work is free from plagiarism.

Thanking You

- Shivani Kumari (2401080001)
- Sadhna Kumari (2401080002)
- Ipsita Mohanty (2401080014)
- Goldi Kumari (2401080041)
- Laxmipriya Sethi (2401080063)
- Barsarani Patra (2401080068)

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At this important stage, we would like to extend our sincere thanks to our instructors whose valuable support helped us in understanding the concepts and successfully completing our project titled **"Loan Approval Prediction."** Their guidance, feedback, and mentorship enabled us to overcome challenges and complete this academic work with confidence.

We would like to express our special gratitude to **Ms. Priya Mishra** and **Ms. Pooja Rani Khuntia (Faculty – Fundamentals of Data Science)** for their valuable guidance and support throughout the project. Their expertise, clear explanations, and practical approach towards teaching helped us understand data science concepts and apply them effectively in our project.

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Thanking You

- Shivani Kumari (2401080001)
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- Barsarani Patra (2401080068)

TABLE OF CONTENT

S No.	Content's	Page No.
01	Abstract	05
02	Introduction	06
03	Problem Statement	07
04	Implementation and Screenshots of Coding	08-13
05	Results	14-17
06	Future Work	18
07	Conclusion	19
08	References	20

ABSTRARCT

The objective of this case study is to develop an intelligent Loan Approval Prediction System using data science and machine learning techniques. In traditional banking systems, loan approval decisions are made manually by analyzing applicant information such as income, credit history, employment status, and financial background. This process is often time-consuming and may lead to inconsistent or inaccurate decisions.

To overcome these limitations, this project implements a machine learning-based approach that analyzes historical loan applicant data to predict whether a loan should be approved or rejected. The dataset includes various attributes such as applicant income, credit score, loan amount, and employment status. Data preprocessing techniques, including data cleaning, handling missing values, and encoding categorical variables, were applied to prepare the dataset for model training.

Multiple machine learning algorithms, including Logistic Regression, Decision Tree, and Random Forest, were implemented and evaluated. Among these models, the Random Forest algorithm achieved the highest accuracy and was selected as the final model for prediction.

Furthermore, a user-friendly web application was developed using Streamlit, allowing users to input their details and receive real-time predictions. The system not only improves the efficiency of loan approval processes but also reduces the risk of default by providing data-driven decisions.
This project demonstrates how machine learning can be effectively applied in the banking sector to automate decision-making, enhance accuracy, and improve overall operational efficiency.

INTRODUCTION

In the modern financial world, banks and lending institutions handle a large number of loan applications on a daily basis. Evaluating these applications is a critical process, as it determines whether an applicant is eligible to receive financial support. Traditionally, this evaluation is performed manually by analyzing various factors such as income level, credit history, employment status, and repayment capability. However, this manual process is time-consuming, prone to human error, and may result in inconsistent decision-making.

With the rapid growth of technology, especially in the field of Data Science and Machine Learning, financial institutions are shifting towards automated systems for better efficiency and accuracy. Machine learning algorithms have the ability to analyze large volumes of data, identify patterns, and make predictions based on historical information. These capabilities make them highly suitable for solving problems related to loan approval and risk assessment.

This project focuses on developing a Loan Approval Prediction System using machine learning techniques. The system analyzes applicant details such as age, income, credit score, loan amount, and employment status to predict whether a loan should be approved or rejected. By training the model on historical data, the system learns patterns associated with successful repayments and high-risk applicants.

To make the system more practical and user-friendly, a web-based application has been developed using Streamlit. This application allows users to input their details and receive instant predictions regarding loan approval. The integration of machine learning with a web interface demonstrates how intelligent systems can assist financial institutions in making faster, more accurate, and data-driven decisions.

Overall, this project highlights the importance of automation in the banking sector and showcases how machine learning can improve efficiency, reduce risk, and enhance the loan approval process.

PROBLEM STATEMENT

Banks and financial institutions receive a large number of loan applications on a daily basis. Evaluating these applications accurately is very important to ensure that loans are granted only to eligible applicants who have the ability to repay them. Traditionally, this process is carried out manually by analyzing various factors such as income, credit score, employment status, and financial history. However, this manual approach is time-consuming and may lead to errors and inconsistent decision-making.

The major challenges in the traditional loan approval system include:

1. **Time-Consuming Process**
   Manual verification of each application takes a lot of time, especially when the number of applicants is high.
2. **Risk of Loan Default**
   Approving loans without proper analysis can increase the chances of non-repayment and financial loss.
3. **Human Errors and Bias**
   Decisions based on human judgment may sometimes be incorrect or biased.
4. **Inconsistent Decision-Making**
   Different evaluators may produce different results for similar applications.
5. **Lack of Automation**
   Traditional systems do not use modern technologies for efficient and fast processing.
6. **Difficulty in Handling Large Data**
   Managing and analyzing large volumes of applicant data manually is difficult and inefficient.

Considering these challenges, there is a need for an intelligent and automated system that can analyze applicant data quickly and provide accurate predictions.

IMPLEMENTATION AND SCREENSHOTS OF CODING

This section explains the step-by-step implementation of the Loan Approval Prediction System using Machine Learning techniques. The system was developed using Python and various data science libraries to perform data analysis, model building, and deployment.

❖ Tools and Technologies Used

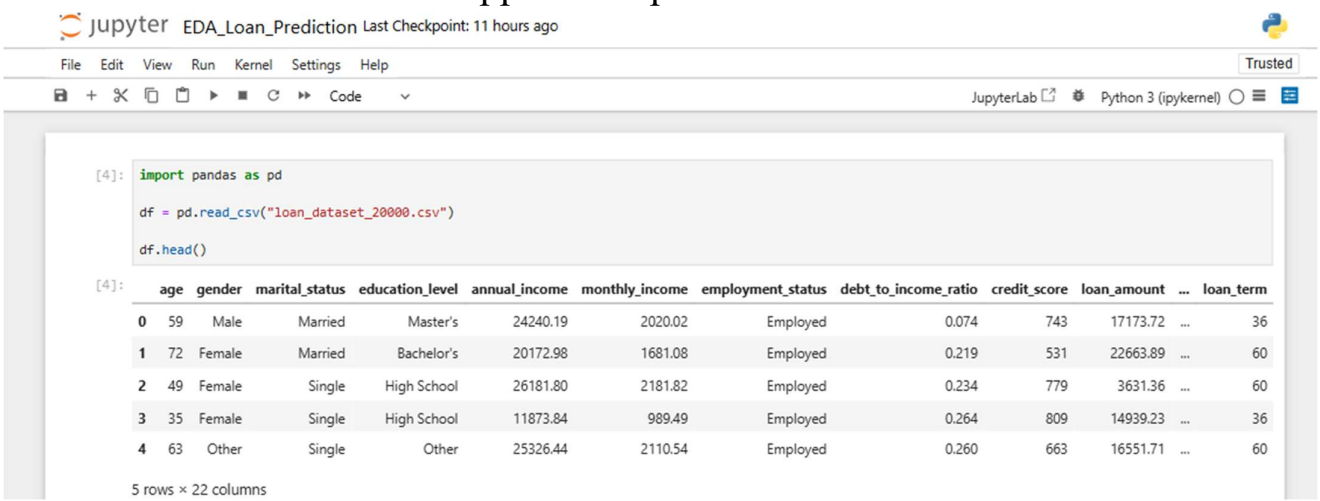
The implementation of the project was carried out using the following tools and libraries:

- **Python** – Main programming language
- **Pandas & NumPy** – Data manipulation and numerical operations
- **Matplotlib & Seaborn** – Data visualization
- **Scikit-learn** – Machine learning model development
- **Streamlit** – Web application development

---

1. Dataset Description

The dataset used in this project contains financial and demographic information of loan applicants. It includes attributes such as age, gender, marital status, education level, employment status, annual income, credit score, loan amount, loan purpose, and other financial indicators. The dataset consists of approximately 20,000 records and multiple features that influence loan repayment behavior. The target variable used for prediction is **loan_paid_back**, which indicates whether the applicant repaid the loan or not.



	age	gender	marital_status	education_level	annual_income	monthly_income	employment_status	debt_to_income_ratio	credit_score	loan_amount	...	loan_term
0	59	Male	Married	Master's	24240.19	2020.02	Employed	0.074	743	17173.72	...	36
1	72	Female	Married	Bachelor's	20172.98	1681.08	Employed	0.219	531	22663.89	...	60
2	49	Female	Single	High School	26181.80	2181.82	Employed	0.234	779	3631.36	...	60
3	35	Female	Single	High School	11873.84	989.49	Employed	0.264	809	14939.23	...	36
4	63	Other	Single	Other	25326.44	2110.54	Employed	0.260	663	16551.71	...	60

2. Data Preprocessing

Before training the Machine Learning model, the dataset was preprocessed to make it suitable for analysis. This step involved checking the dataset structure, identifying missing values, and converting categorical data into numerical format. Categorical variables such as gender, marital status, education level, and loan purpose were converted using One-Hot Encoding. This process ensured that the dataset contained only numerical values that could be processed by Machine Learning algorithms.

```
[9]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20000 entries, 0 to 19999
Data columns (total 22 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   age                   20000 non-null  int64
 1   gender                20000 non-null  object
 2   marital_status        20000 non-null  object
 3   education_level       20000 non-null  object
 4   annual_income         20000 non-null  float64
 5   monthly_income        20000 non-null  float64
 6   employment_status     20000 non-null  object
 7   debt_to_income_ratio  20000 non-null  float64
 8   credit_score          20000 non-null  int64
 9   loan_amount           20000 non-null  float64
 10  loan_purpose          20000 non-null  object
 11  interest_rate         20000 non-null  float64
 12  loan_term             20000 non-null  int64
 13  installment           20000 non-null  float64
 14  grade_subgrade        20000 non-null  object
 15  num_of_open_accounts  20000 non-null  int64
 16  total_credit_limit    20000 non-null  float64
 17  current_balance       20000 non-null  float64
 18  delinquency_history   20000 non-null  int64
 19  public_records        20000 non-null  int64
 20  num_of_delinquencies  20000 non-null  int64
 21  loan_paid_back        20000 non-null  int64
dtypes: float64(8), int64(8), object(6)
memory usage: 3.4+ MB
```

3. Feature Selection and Data Splitting

After preprocessing, the features and target variable were separated. The feature set (X) contained all independent variables such as income, credit score, and loan amount, while the target variable (y) represented the loan repayment status. The dataset was then divided into training and testing sets using the train_test_split function. Approximately 80% of the data was used for training the model, and 20% was used for testing its performance.

```
[16]: categorical_cols = df.select_dtypes(include=['object']).columns

print(categorical_cols)
```

```
Index(['gender', 'marital_status', 'education_level', 'employment_status',
       'loan_purpose', 'grade_subgrade'],
      dtype='object')
```

```
[17]: df_encoded = pd.get_dummies(df, columns=categorical_cols, drop_first=True)

df_encoded.head()
```

[17]:

	age	annual_income	monthly_income	debt_to_income_ratio	credit_score	loan_amount	interest_rate	loan_term	installment	num_of_open_accounts	...	grade_sub
0	59	24240.19	2020.02	0.074	743	17173.72	13.39	36	581.88	7	...	
1	72	20172.98	1681.08	0.219	531	22663.89	17.81	60	573.17	5	...	
2	49	26181.80	2181.82	0.234	779	3631.36	9.53	60	76.32	2	...	
3	35	11873.84	989.49	0.264	809	14939.23	7.99	36	468.07	7	...	
4	63	25326.44	2110.54	0.260	663	16551.71	15.20	60	395.50	1	...	

5 rows × 65 columns

```
[18]: X = df_encoded.drop("loan_paid_back", axis=1)

y = df_encoded["loan_paid_back"]
```

```
[19]: from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
[20]: print("Training Data:", X_train.shape)
print("Testing Data:", X_test.shape)
```

```
Training Data: (16000, 64)
Testing Data: (4000, 64)
```

4. Model Training

Multiple Machine Learning algorithms were implemented to evaluate their performance in predicting loan repayment. The models used in this project include:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier

Each model was trained using the training dataset, and predictions were made on the testing dataset.

```
[21]: from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

log_model = LogisticRegression(max_iter=1000)

log_model.fit(X_train, y_train)
```

```
[22]: from sklearn.tree import DecisionTreeClassifier

dt_model = DecisionTreeClassifier()

dt_model.fit(X_train, y_train)

dt_pred = dt_model.predict(X_test)

dt_acc = accuracy_score(y_test, dt_pred)

print("Decision Tree Accuracy:", dt_acc)
```

```
Decision Tree Accuracy: 0.83325
```

```
[23]: from sklearn.ensemble import RandomForestClassifier

rf_model = RandomForestClassifier()

rf_model.fit(X_train, y_train)

rf_pred = rf_model.predict(X_test)

rf_acc = accuracy_score(y_test, rf_pred)

print("Random Forest Accuracy:", rf_acc)
```

```
Random Forest Accuracy: 0.8935
```

5. Model Evaluation

The performance of each model was evaluated using accuracy metrics. The results showed that the Random Forest Classifier provided the highest prediction accuracy compared to the other models. Therefore, Random Forest was selected as the final model for the loan approval prediction system.

```
[24]: print("Logistic Regression:", log_acc)
      print("Decision Tree:", dt_acc)
      print("Random Forest:", rf_acc)

      Logistic Regression: 0.79175
      Decision Tree: 0.83325
      Random Forest: 0.8935
```

6. Model Saving

After selecting the best-performing model, it was saved using the Pickle library so that it could be used later in the web application without retraining the model.

```
[28]: import pickle

      pickle.dump(rf_model, open("loan_model.pkl", "wb"))
      feature_columns = X.columns

      pickle.dump(feature_columns, open("model_features.pkl", "wb"))
```

7. Development of Streamlit Web Application

To make the system interactive and user-friendly, a web application was developed using the Streamlit framework. Streamlit allows Machine Learning models to be integrated into a simple web interface where users can enter their information and receive instant predictions. In this project, the trained Random Forest model was connected with a Streamlit application so that the prediction system can be used easily through a web-based interface.

The application was implemented using a Python script named **app.py**. In this script, the trained Machine Learning model was first loaded using the Pickle library. After loading the model, the Streamlit library was used to create the layout and structure of the web application.

app.py

app.py > ...

```python
import streamlit as st
import pickle
import pandas as pd

# ================= LOAD MODEL =================
model = pickle.load(open("loan_model.pkl", "rb"))
features = pickle.load(open("model_features.pkl", "rb"))

st.set_page_config(page_title="CGU Bank Loan System", layout="wide")

# ================= SIDEBAR =================

st.sidebar.image("logo.png", width=150)

st.sidebar.title("CGU Bank")

```

Next, the application provides input fields where users can enter important loan-related details such as age, annual income, credit score, and loan amount. These input fields are created using Streamlit form components such as number inputs, text inputs, and selection boxes.

```python
# ================ PREDICTION ================

if st.button("Predict Loan Approval"):

    input_data = {
        "age": age,
        "annual_income": annual_income,
        "credit_score": credit_score,
        "loan_amount": loan_amount
    }

    input_df = pd.DataFrame([input_data])

    for col in features:
        if col not in input_df:
            input_df[col] = 0

    input_df = input_df[features]

    prediction = model.predict(input_df)
    probability = model.predict_proba(input_df)[0][1] * 100

    st.subheader("Loan Prediction Result")

    st.metric("Approval Probability", f"{probability:.2f}%")
```

After the user enters the required information and clicks the prediction button, the entered data is processed and sent to the trained Machine Learning model. The model then analyzes the input values and predicts whether the loan should be approved or rejected.

```python
if prediction[0] == 1:

    st.success("✅ Loan Approved")

    st.info(f"""
    Congratulations **{name}**!

    Based on your financial profile, our AI system predicts that you are **likely to repay the loan successfully**.

    Your application has **low financial risk** and meets CGU Bank's lending criteria.
    """)

else:

    st.error("❌ Loan Rejected")

    st.warning(f"""
    Dear **{name}**,

    Our system detected **high financial risk** based on the provided details.

    Suggestions to improve eligibility:
    • Improve credit score
    • Reduce existing debt
    • Increase annual income
    """)

st.divider()

st.caption("CGU Bank AI Loan Risk Analysis System | Developed for BCA 6th Semester ML Case Study")
```

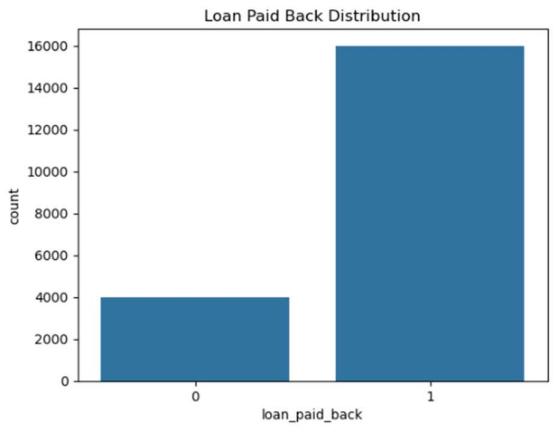
This Streamlit-based implementation demonstrates how Machine Learning models can be deployed through a web application to create an interactive and intelligent decision-support system for loan approval prediction.

RESULTS

This section presents the results obtained from the data analysis, model evaluation, and the final prediction system developed in this project. Several visualizations were created to better understand the dataset and the performance of the Data Science & Machine Learning model.

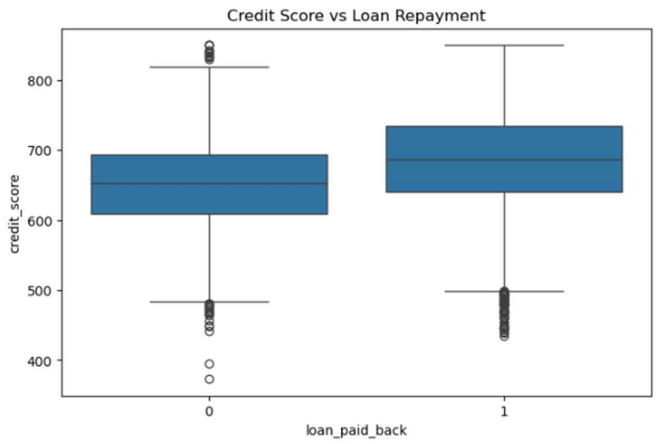
First, a distribution analysis was performed to observe the number of applicants who repaid their loan and those who did not. This visualization helps in understanding the balance between the two classes in the dataset.

```
[11]: import seaborn as sns
      import matplotlib.pyplot as plt

      sns.countplot(x='loan_paid_back', data=df)

      plt.title("Loan Paid Back Distribution")

      plt.show()
```



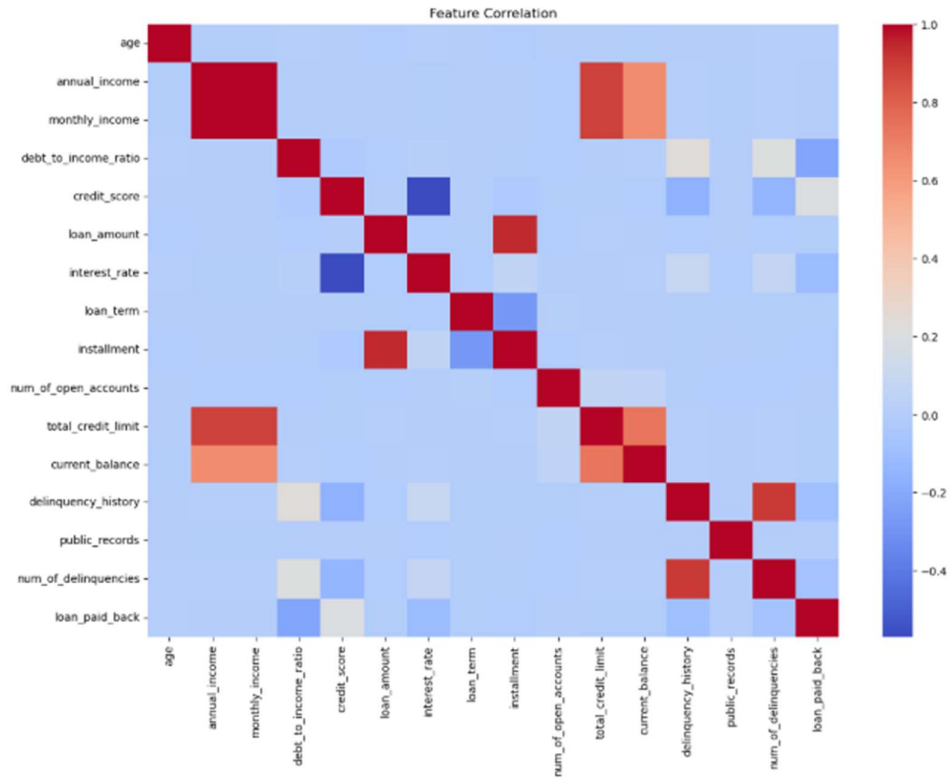
Next, a boxplot was used to analyze the relationship between the applicant's credit score and loan repayment status. This visualization helps identify how credit score influences the likelihood of loan repayment.

```
[12]: plt.figure(figsize=(8,5))

      sns.boxplot(x='loan_paid_back', y='credit_score', data=df)

      plt.title("Credit Score vs Loan Repayment")

      plt.show()
```



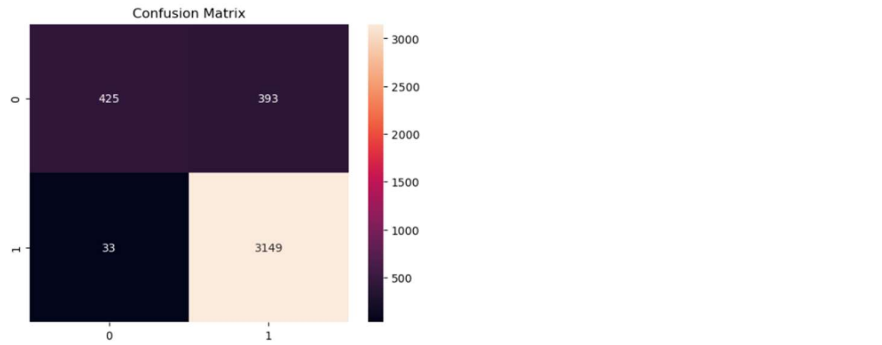
A correlation heatmap was also generated to understand the relationships between different numerical features in the dataset. This heatmap highlights how strongly different variables are related to each other and helps identify important features that influence loan repayment.

```
[15]: plt.figure(figsize=(14,10))
      sns.heatmap(df.corr(numeric_only=True), cmap="coolwarm")
      plt.title("Feature Correlation")
      plt.show()
```

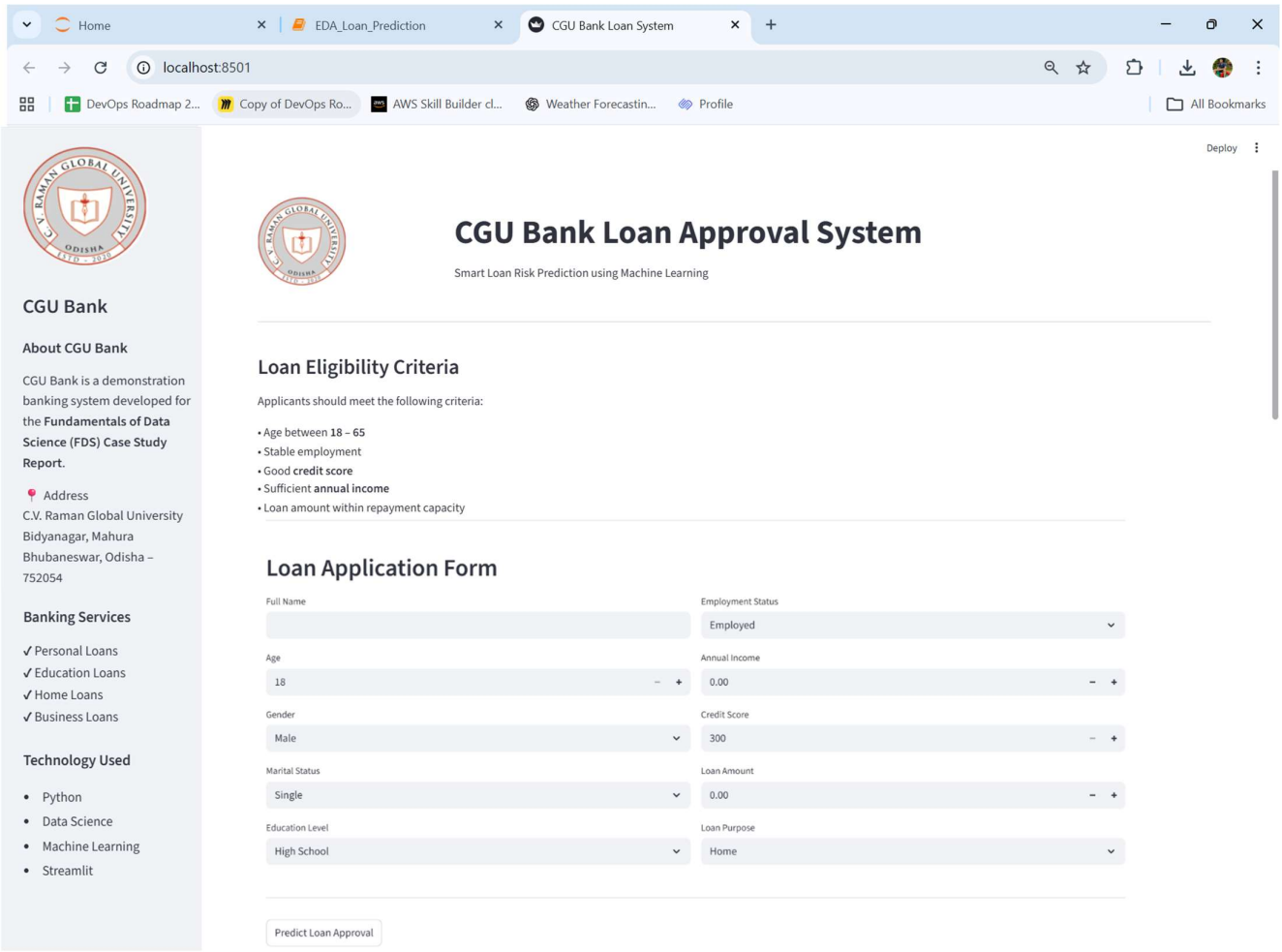


To evaluate the performance of the final model, a confusion matrix was generated using the Random Forest classifier. The confusion matrix shows the number of correct and incorrect predictions made by the model, which helps assess the model's classification performance.

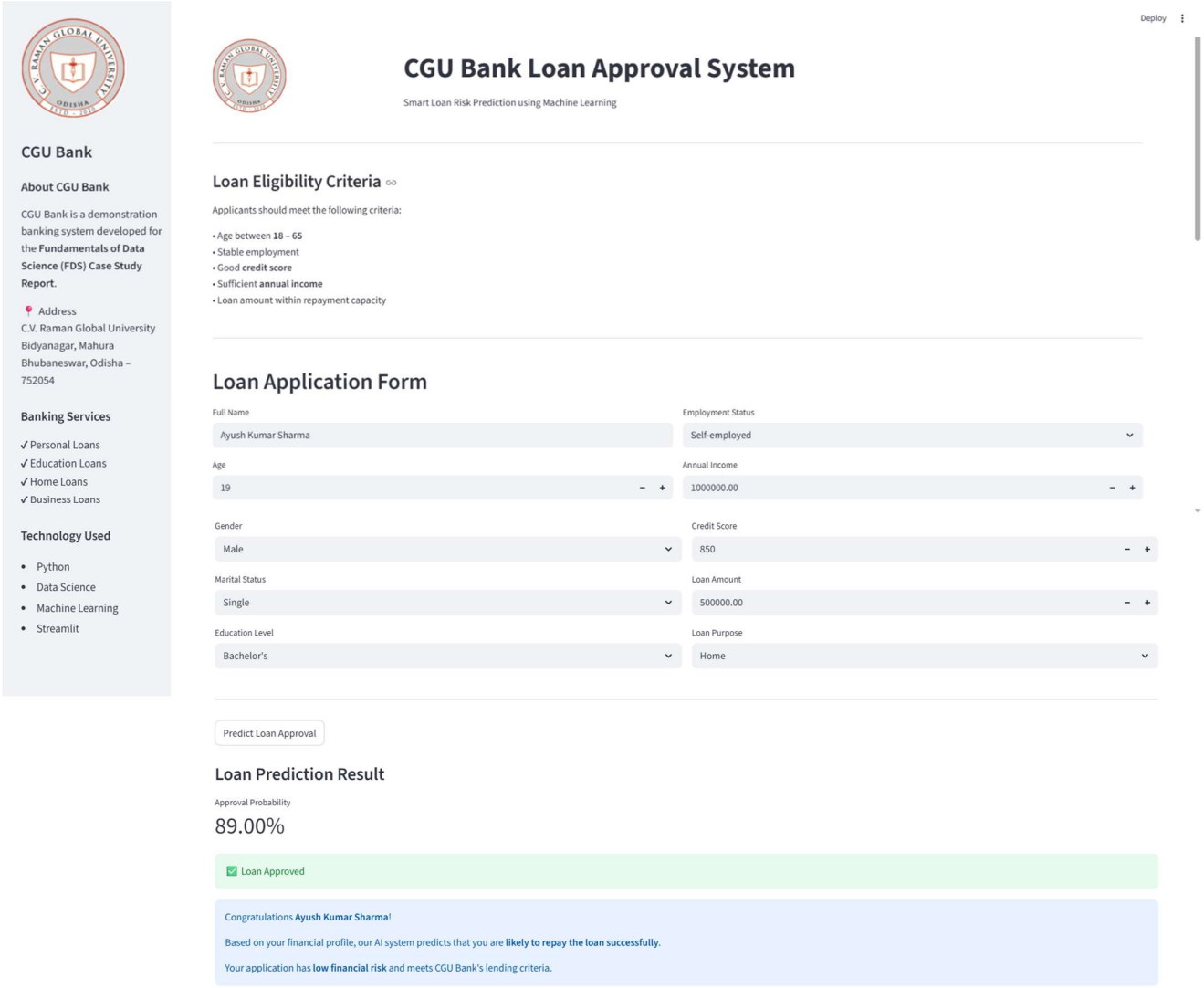
```
[25]: from sklearn.metrics import confusion_matrix
      cm = confusion_matrix(y_test, rf_pred)
      sns.heatmap(cm, annot=True, fmt='d')
      plt.title("Confusion Matrix")
      plt.show()
```



After evaluating the model performance, the trained model was integrated into a Streamlit-based web application. The web application provides an interface where users can enter loan applicant details such as age, income, credit score, and loan amount.



Once the user enters the required details and clicks the prediction button, the system processes the input data using the trained Machine Learning model and displays the prediction result, indicating whether the loan is likely to be approved or rejected along with the probability of approval.



The results indicate that the developed data-driven system can efficiently analyze applicant details and generate accurate loan approval predictions through an interactive web-based application.

FUTURE WORK

Although the Loan Approval Prediction System developed in this project performs effectively, there are several opportunities for further improvement and enhancement. With advancements in technology and availability of larger datasets, the system can be made more accurate, efficient, and suitable for real-world applications. Future developments can focus on improving model performance, expanding features, and integrating the system with real-time environments.

The possible future enhancements of this project include:

1. **Use of Advanced Algorithms**
   Implementing advanced machine learning models such as XGBoost, Gradient Boosting, and Neural Networks to improve prediction accuracy.
2. **Integration with Real-Time Systems**
   Connecting the model with real-time banking databases to make live loan approval decisions.
3. **Larger and More Diverse Dataset**
   Using a bigger and more diverse dataset to improve model training and generalization.
4. **Improved Feature Engineering**
   Adding more relevant features such as transaction history and credit behavior for better prediction.
5. **Model Optimization**
   Applying hyperparameter tuning techniques to enhance model performance.
6. **Security and Data Privacy**
   Implementing secure systems to protect sensitive user and financial data.
7. **Mobile Application Development**
   Developing a mobile-based application for easier accessibility and wider usage.

CONCLUSION

This case study successfully demonstrates the development of a Loan Approval Prediction System using data science and machine learning techniques. The system is designed to analyze applicant information and predict whether a loan should be approved or rejected based on various factors such as income, credit score, employment status, and loan amount.

The project involved multiple stages, including data preprocessing, feature selection, model training, and performance evaluation. Different machine learning algorithms such as Logistic Regression, Decision Tree, and Random Forest were implemented and compared. Among these, the Random Forest Classifier provided the best performance in terms of accuracy and reliability and was selected as the final model.

Furthermore, the integration of the trained model into a Streamlit-based web application made the system user-friendly and interactive. Users can easily input their details and receive instant predictions, making the system practical for real-world applications.

The developed system highlights the importance of automation in the banking sector. It reduces manual effort, minimizes human errors, and improves the speed and accuracy of loan approval decisions. Additionally, it helps financial institutions in identifying potential risks and making better data-driven decisions. In conclusion, this project proves that machine learning can play a significant role in transforming traditional loan approval processes into efficient, reliable, and intelligent systems.

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